

Quantum error correction

12 notes · Siirsuite

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The information we want to preserve

01.01 · An unknown quantum state

EXPLANATION

Preserving a qubit means preserving its whole state. Its populations determine computational-basis measurement probabilities; its coherence carries the relative phase that affects other measurements. Measuring in the computational basis and keeping the answer preserves only part of this information.

The off-diagonal entries of the density matrix make the missing information explicit. Removing them can change the quantum state even when the probabilities of zero and one stay unchanged.

MATHS

$$\mathcal{H}_L = \mathbb{C}^2, \quad |\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

$$\rho_\psi = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}.$$

$$\Delta_Z(\rho) = \sum_{b=0}^1 |b\rangle\langle b|\rho|b\rangle\langle b|, \quad \Delta_Z(\rho_\psi) = \begin{pmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{pmatrix}.$$

01.02 · Preservation includes external entanglement

EXPLANATION

The stored qubit may be entangled with another system. A successful memory must preserve those correlations as well. The reference system in the equation represents anything outside the device; recovery acts only on the encoded qubit.

For example, one half of a Bell pair looks individually maximally mixed both before and after dephasing. Yet the joint state has lost its entanglement. Checking only the local populations would miss this failure.

MATHS

$$\mathcal{V}(\rho) = V\rho V^\dagger, \quad \mathcal{N}, \mathcal{R} : \mathcal{B}(\mathcal{H}_P) \rightarrow \mathcal{B}(\mathcal{H}_P) \quad \text{CPTP}.$$

Exact recovery:

$$\forall \rho_{RL}, \quad (\text{id}_R \otimes \mathcal{R} \circ \mathcal{N} \circ \mathcal{V})(\rho_{RL}) = (\text{id}_R \otimes \mathcal{V})(\rho_{RL}).$$

Witness state:

$$|\Phi^+\rangle_{RL} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad (\text{id} \otimes \Delta_Z)(|\Phi^+\rangle\langle\Phi^+|) = \frac{|00\rangle\langle 00| + |11\rangle\langle 11|}{2}.$$

01.03 · No-cloning

EXPLANATION

A unitary preserves inner products. If it copied every state, the overlap between two possible inputs would have to equal the square of that overlap. Distinct nonorthogonal states violate that requirement.

A device can copy the labels of orthogonal basis states. Applied to a superposition, that same device creates correlations between its outputs. Linearity determines this behavior: it does not create two independent copies of the unknown input.

MATHS

Suppose, for all normalized $|u\rangle, |v\rangle$,

$$U|u\rangle|0\rangle = |u\rangle|u\rangle, \quad U|v\rangle|0\rangle = |v\rangle|v\rangle.$$

Unitarity:

$$a := \langle u|v\rangle = \langle u, u|v, v\rangle = a^2 \implies a \in \{0, 1\}.$$

$$0 < |\langle u|v\rangle| < 1 \implies \text{contradiction.}$$

For $U|0, 0\rangle = |00\rangle$ and $U|1, 0\rangle = |11\rangle$:

$$U|\psi, 0\rangle = \alpha|00\rangle + \beta|11\rangle,$$

$$|\psi\rangle^{\otimes 2} = \alpha^2|00\rangle + \alpha\beta(|01\rangle + |10\rangle) + \beta^2|11\rangle.$$

01.04 · Encoding through correlations

EXPLANATION

The repetition encoding places the amplitudes in two collective states of three qubits. Each individual physical qubit has lost access to the original relative phase; the full encoded system still carries it.

This is the redundancy quantum error correction can use. Correlations let us ask whether qubits agree without asking which logical value the state contains. Encoding a plus state gives a GHZ state.

MATHS

$$V|0\rangle = |000\rangle =: |0_L\rangle, \quad V|1\rangle = |111\rangle =: |1_L\rangle.$$

$$V|\psi\rangle = \text{CNOT}_{1\rightarrow 3} \text{CNOT}_{1\rightarrow 2} (|\psi\rangle_1 |00\rangle_{23}) = \alpha|000\rangle + \beta|111\rangle.$$

$$\text{Tr}_{\{1,2,3\}\setminus\{j\}}(V\rho_\psi V^\dagger) = |\alpha|^2|0\rangle\langle 0| + |\beta|^2|1\rangle\langle 1|, \quad j = 1, 2, 3.$$

$$V|+\rangle = (|000\rangle + |111\rangle)/\sqrt{2}.$$

01.05 · What this first code protects

EXPLANATION

The two parity checks distinguish no bit flip from a bit flip on each of the three positions. Under that restricted error model, the syndrome tells us how to repair the state.

A phase flip on one qubit is different: it changes the logical relative phase while leaving both parities unchanged. Thus this code corrects a single bit flip, but it does not correct arbitrary single-qubit noise. Its quantum distance is only one.

MATHS

$$\mathcal{E}_X = \{I, X_1, X_2, X_3\}, \quad g_1 = Z_1 Z_2, \quad g_2 = Z_2 Z_3.$$

E	I	X_1	X_2	X_3
(s_1, s_2)	00	10	11	01

$$\bar{Z} = Z_1, \quad Z_1 V|\psi\rangle = \alpha|000\rangle - \beta|111\rangle, \quad s(Z_1) = 00.$$

$$\bar{X} = X_1 X_2 X_3, \quad [[n, k, d]] = [[3, 1, 1]].$$

01.06 · Exercise and result

EXPLANATION

Ask why the plus-phase and minus-phase encoded states have identical computational-basis probabilities. Those probabilities depend on squared amplitude magnitudes, which ignore the sign.

The triple-X measurement can distinguish their coherence when the relevant real part is nonzero. For the encoded plus and minus states, its outcomes are opposite and deterministic. Measuring every data qubit in the computational basis destroys this distinction; measuring only the stabilizer parities preserves it.

MATHS

$$|\psi_\pm\rangle = \alpha|000\rangle \pm \beta|111\rangle.$$

$$\Pr_\pm(000) = |\alpha|^2, \quad \Pr_\pm(111) = |\beta|^2, \quad \Pr_\pm(b) = 0 \quad (b \notin \{000, 111\}).$$

$$\langle \psi_\pm | X_1 X_2 X_3 | \psi_\pm \rangle = \pm 2 \text{Re}(\alpha^* \beta).$$

Encoding into a code space

02.01 · Encoding isometry

EXPLANATION

An encoding selects a small subspace inside a larger physical Hilbert space. The isometry preserves inner products, so it preserves all distinctions among logical states.

The code space contains every allowed logical superposition. Its dimension is determined by the number of logical qubits. Adding physical qubits enlarges the surrounding space without increasing the amount of logical information being stored.

MATHS

$$\begin{aligned}\mathcal{H}_L &= (\mathbb{C}^2)^{\otimes k}, \quad \mathcal{H}_P = (\mathbb{C}^2)^{\otimes n}, \quad n \geq k, \\ V : \mathcal{H}_L &\rightarrow \mathcal{H}_P, \quad V^\dagger V = I_L, \quad \mathcal{C} = \text{im } V, \quad \dim \mathcal{C} = 2^k. \\ \langle V\phi | V\psi \rangle &= \langle \phi | \psi \rangle, \quad V \sum_{j=0}^{2^k-1} \alpha_j |j\rangle = \sum_{j=0}^{2^k-1} \alpha_j |j_L\rangle.\end{aligned}$$

02.02 · Code projector

EXPLANATION

The projector describes membership in the code space without choosing a logical value. Applied to a valid codeword, it leaves the state unchanged.

An ideal yes-or-no measurement of this projector asks whether the system lies in the code space. It does not reveal which superposition within that space is present. Later, several commuting checks refine this question into a full syndrome.

MATHS

$$\begin{aligned}P &:= VV^\dagger = \sum_{j=0}^{2^k-1} |j_L\rangle\langle j_L|, \\ P^\dagger &= P, \quad P^2 = V(V^\dagger V)V^\dagger = P, \quad \text{rank } P = \text{Tr } P = 2^k. \\ |\phi\rangle \in \mathcal{C} &\iff P|\phi\rangle = |\phi\rangle, \quad \rho \text{ supported on } \mathcal{C} \iff P\rho P = \rho.\end{aligned}$$

$$\Pr(\mathcal{C} \mid \rho) = \text{Tr}(P\rho).$$

02.03 · Repetition-code sectors

EXPLANATION

A bit flip moves the encoded state into another two-dimensional subspace. The original amplitudes remain attached to the two basis vectors in that new sector.

These four sectors are mutually orthogonal and fill the physical space. A measurement can identify the sector while leaving the unknown amplitudes intact. Applying the corresponding inverse bit flip then returns the state to the code space.

MATHS

$$\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}, \quad P = |000\rangle\langle 000| + |111\rangle\langle 111|.$$

$$X_1\mathcal{C} = \text{span}\{|100\rangle, |011\rangle\},$$

$$X_2\mathcal{C} = \text{span}\{|010\rangle, |101\rangle\},$$

$$X_3\mathcal{C} = \text{span}\{|001\rangle, |110\rangle\}.$$

$$\mathcal{H}_P = \mathcal{C} \oplus X_1\mathcal{C} \oplus X_2\mathcal{C} \oplus X_3\mathcal{C}.$$

For $E_a, E_b \in \{I, X_1, X_2, X_3\}$:

$$PE_a^\dagger E_b P = \delta_{ab} P.$$

02.04 · Three kinds of error ambiguity

EXPLANATION

Two physical errors can act identically on every codeword. They can also reach the same subspace while acting differently on the logical state. Those are different forms of ambiguity.

The two-qubit Z stabilizer of the repetition code is harmless, whereas a single Z acts as a logical phase flip. Both leave the state inside the original code space.

A general coherent error can put the state into a superposition of syndrome sectors. The tidy picture of one error moving the state into one sector applies directly to Pauli errors; it is not a catalogue of every possible physical noise process.

MATHS

Same action on the entire code, up to a common phase:

$$FV = e^{i\theta} EV \iff E^\dagger FV = e^{i\theta} V \quad (E, F \text{ unitary}).$$

Same image subspace:

$$EC = FC \not\Rightarrow FV = e^{i\theta} EV.$$

Examples:

$$(Z_1 Z_2)V = V, \quad Z_1 V = V Z_1, \quad Z_1 \mathcal{C} = \mathcal{C}.$$

Coherent error:

$$e^{-i\theta X_1/2} V |\psi\rangle = \cos(\theta/2) V |\psi\rangle - i \sin(\theta/2) X_1 V |\psi\rangle.$$

02.05 · Exercise and result

EXPLANATION

Apply the middle-qubit bit flip to an arbitrary encoded state, then take its inner product with an arbitrary original codeword. Every computational basis overlap vanishes.

This proves orthogonality of the whole displaced subspace to the code space. It does not depend on choosing special amplitudes. The projector onto the displaced sector is obtained by conjugating the original projector by the bit flip.

MATHS

$$X_2 V |\psi\rangle = \alpha |010\rangle + \beta |101\rangle.$$

$$\forall |\phi_L\rangle = \gamma |000\rangle + \delta |111\rangle, \quad \langle \phi_L | X_2 V |\psi\rangle = 0.$$

$$P X_2 P = 0, \quad Q_2 := X_2 P X_2, \quad P Q_2 = 0, \quad Q_2^2 = Q_2.$$

Stabilizers and safe questions

03.01 · Stabilizer group

EXPLANATION

Stabilizers are compatible questions whose answers are fixed for every valid logical state. Choose independent commuting Hermitian Pauli operators, each with required answer plus one.

Commutation makes the joint eigenvalue conditions compatible. Independence prevents counting the same constraint twice. Excluding minus identity prevents a condition that no nonzero state could satisfy. Anticommuting checks cannot both have a nonzero common plus-one eigenvector.

MATHS

$$\mathcal{P}_n = \{i^q P_1 \otimes \cdots \otimes P_n : q \in \mathbb{Z}_4, P_j \in \{I, X, Y, Z\}\}.$$

$$\mathcal{S} = \langle g_1, \dots, g_r \rangle \subset \mathcal{P}_n, \quad g_i^\dagger = g_i, \quad g_i^2 = I, \quad [g_i, g_j] = 0, \quad -I \notin \mathcal{S}.$$

Independence:

$$\prod_{i=1}^r g_i^{u_i} = I \iff u = 0, \quad u \in \mathbb{F}_2^r.$$

$$|\mathcal{S}| = 2^r, \quad \mathcal{C} = \bigcap_{i=1}^r \ker(g_i - I), \quad k = n - r.$$

If $AB = -BA$ and $A|\phi\rangle = B|\phi\rangle = |\phi\rangle$:

$$|\phi\rangle = AB|\phi\rangle = -BA|\phi\rangle = -|\phi\rangle \implies |\phi\rangle = 0.$$

03.02 · Syndrome projectors and dimension

EXPLANATION

A syndrome is the list of check outcomes, with zero representing plus one and one representing minus one. Each projector selects one joint eigenspace.

Independent checks divide the physical space into equally sized orthogonal sectors. The all-zero sector is the code space. In the projector expansion, every nonidentity Pauli has zero trace, which gives the dimension directly.

If a measurement schedule includes redundant checks, their outcomes satisfy constraints; the count of independent generators is what determines the dimension.

MATHS

$$s \in \mathbb{F}_2^r, \quad P_s = \prod_{i=1}^r \frac{I + (-1)^{s_i} g_i}{2}.$$

$$P_s^\dagger = P_s, \quad P_s P_t = \delta_{st} P_s, \quad \sum_{s \in \mathbb{F}_2^r} P_s = I.$$

$$P_0 = 2^{-r} \sum_{g \in \mathcal{S}} g, \quad \text{Tr } P_s = 2^{-r} \text{Tr } I = 2^{n-r} = 2^k.$$

$$\mathcal{H}_P = \bigoplus_{s \in \mathbb{F}_2^r} \text{im } P_s.$$

03.03 · Ideal measurement preserves internal coherence

EXPLANATION

An ideal stabilizer measurement learns which sector contains the state. It does not distinguish logical amplitudes within that sector. A codeword therefore survives the measurement unchanged.

For a fixed Pauli error, the syndrome is determined by the error and is independent of the encoded amplitudes. This statement assumes an ideal check measurement. Faults in the ancilla circuit must be analyzed separately.

MATHS

$$\mathcal{M}_s(\rho) = P_s \rho P_s, \quad p_s = \text{Tr}(P_s \rho), \quad \rho_s = P_s \rho P_s / p_s \quad (p_s > 0).$$

$$P_0 \rho_L P_0 = \rho_L \implies p_0 = 1, \quad \rho_0 = \rho_L.$$

For Pauli E with syndrome $s(E)$:

$$P_s E V = \delta_{s, s(E)} E V.$$

03.04 · Ancilla measurement of Z parity

EXPLANATION

Prepare the ancilla in zero. Each data qubit controls a CNOT into it. The ancilla flips once for each data one, so it stores only the parity.

Both components of an even-parity superposition leave the ancilla in the same state. Measuring that ancilla therefore does not reveal which even-parity component was present. Measuring the two data qubits individually would reveal more information and destroy their superposition.

MATHS

$$U_Z = \text{CNOT}_{2 \rightarrow a} \text{CNOT}_{1 \rightarrow a}, \quad U_Z |b_1 b_2\rangle |0\rangle_a = |b_1 b_2\rangle |b_1 \oplus b_2\rangle_a.$$

$$K_m := {}_a \langle m | U_Z | 0 \rangle_a = \frac{I + (-1)^m Z_1 Z_2}{2}, \quad m \in \{0, 1\}.$$

$$U_Z(\alpha|00\rangle + \beta|11\rangle)|0\rangle_a = (\alpha|00\rangle + \beta|11\rangle)|0\rangle_a.$$

03.05 · Ancilla measurement of X parity

EXPLANATION

For an X check, prepare the ancilla in plus and use it as the control of CNOTs into the data qubits. Measure the ancilla in the X basis.

The two possible results project the data onto the two eigenspaces of the X product. The ancilla direction matters: this circuit is the complementary construction to the Z-parity circuit. Its ideal action does not by itself establish fault tolerance.

MATHS

$$A = \prod_{j \in Q} X_j, \quad U_X = |0\rangle\langle 0|_a \otimes I + |1\rangle\langle 1|_a \otimes A.$$

$$|\pm_m\rangle = \frac{|0\rangle + (-1)^m |1\rangle}{\sqrt{2}}, \quad K_m = {}_a \langle \pm_m | U_X | + \rangle_a = \frac{I + (-1)^m A}{2}.$$

$$U_X = \prod_{j \in Q} \text{CNOT}_{a \rightarrow j}.$$

03.06 · Exercise and result

EXPLANATION

Both repetition-code parity checks return plus one on both logical basis states. Their outcomes therefore cannot distinguish the logical basis values.

Measuring the first data qubit in Z does distinguish them. For a general encoded superposition it selects one logical branch. The difference is the information the question reveals, even though all three observables are built from Z operators.

$$g_1 = Z_1 Z_2, \quad g_2 = Z_2 Z_3, \quad g_i |b_L\rangle = |b_L\rangle \quad (i = 1, 2; b = 0, 1).$$

$$Z_1 |0_L\rangle = |0_L\rangle, \quad Z_1 |1_L\rangle = -|1_L\rangle.$$

$$\Pi_{\pm} = (I \pm Z_1)/2, \quad \Pi_+ |\psi_L\rangle = \alpha |0_L\rangle, \quad \Pi_- |\psi_L\rangle = \beta |1_L\rangle.$$

Errors become syndrome patterns

04.01 · Pauli expansion and correctable spans

EXPLANATION

Pauli operators form a basis for all operators, so they give us a finite language for describing continuous physical errors. The correction condition must hold for every pair of errors in the chosen set. When it does, it also holds for linear combinations of those errors.

This does not turn every channel into a classical random Pauli. A general channel contains cross terms between Pauli components. Such coherent terms disappear only under additional assumptions or an explicit approximation.

MATHS

$$\mathcal{B}(\mathcal{H}_P) = \text{span}_{\mathbb{C}}\{I, X, Y, Z\}^{\otimes n}, \quad A = 2^{-n} \sum_Q \text{Tr}(Q^\dagger A) Q.$$

$$P E_a^\dagger E_b P = c_{ab} P, \quad F_\mu = \sum_a f_{\mu a} E_a \implies P F_\mu^\dagger F_\nu P = \left(\sum_{a,b} f_{\mu a}^* c_{ab} f_{\nu b} \right) P.$$

$$\mathcal{N}(\rho) = \sum_\mu A_\mu \rho A_\mu^\dagger, \quad A_\mu = \sum_Q a_{\mu Q} Q, \quad \mathcal{N}(\rho) = \sum_{Q,Q'} \chi_{QQ'} Q \rho Q'^\dagger.$$

Pauli mixture: $\chi_{QQ'} = 0$ for $Q \neq Q'$

04.02 · Anticommutation produces the syndrome

EXPLANATION

A Pauli error flips exactly the checks it anticommutes with. The derivation simply moves the check past the error and then uses the fact that the original state was stabilized.

Composing two Pauli errors adds their syndrome bits modulo two. A check flipped twice returns to its original sign. This cancellation is the algebra behind the endpoints of error strings.

MATHS

$$g_i E = (-1)^{s_i(E)} E g_i, \quad s_i(E) \in \mathbb{F}_2.$$

$$g_i |\psi_L\rangle = |\psi_L\rangle \implies g_i E |\psi_L\rangle = (-1)^{s_i(E)} E |\psi_L\rangle.$$

$$s(EF) = s(E) + s(F) \pmod{2}.$$

04.03 · Binary symplectic representation

EXPLANATION

Represent each qubit by two bits recording its X and Z components. A Y error has both bits set. We ignore overall Pauli phase in this representation because it does not affect a standalone error's syndrome or logical class.

The symplectic matrix swaps the two halves so that X components are compared against Z components. One matrix multiplication then computes the syndrome. The check-commutation condition uses the same pairing.

MATHS

$$E \sim X^{\mathbf{x}} Z^{\mathbf{z}}, \quad \mathbf{x}, \mathbf{z} \in \mathbb{F}_2^n, \quad e = \begin{pmatrix} \mathbf{x} \\ \mathbf{z} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}.$$

$$E(e)E(f) = (-1)^{e^T J f} E(f)E(e).$$

For $g_i \sim X^{\mathbf{a}_i} Z^{\mathbf{b}_i}$:

$$H = \begin{pmatrix} \mathbf{a}_1^T & \mathbf{b}_1^T \\ \vdots & \vdots \\ \mathbf{a}_r^T & \mathbf{b}_r^T \end{pmatrix}, \quad s = H J e, \quad H J H^T = 0.$$

$$(x_j, z_j) : I \leftrightarrow (0, 0), \quad X \leftrightarrow (1, 0), \quad Z \leftrightarrow (0, 1), \quad Y \leftrightarrow (1, 1).$$

04.04 · CSS specialization

EXPLANATION

CSS codes separate the checks into pure X and pure Z products. X checks detect the Z component of an error, and Z checks detect its X component.

The subscripts on syndrome vectors here name the measured check type. Keeping that convention explicit prevents a common mix-up when building two decoding graphs. The equation relating the check matrices says every X check overlaps every Z check on an even number of qubits.

MATHS

$$g_a^X = X^{(H_X)_{a,:}}, \quad g_b^Z = Z^{(H_Z)_{b,:}}, \quad H_X H_Z^T = 0.$$

$$s_X = H_X \mathbf{z}, \quad s_Z = H_Z \mathbf{x}, \quad k = n - \text{rank } H_X - \text{rank } H_Z.$$

04.05 · Noise assumptions

EXPLANATION

A noise parameter needs a convention. Here depolarizing probability means the total probability of a nonidentity Pauli, divided equally among X, Y, and Z.

Even if different qubits are independent, the X and Z components on the same qubit generally are not: Y contributes to both. Independent decoding of those components can therefore discard useful information.

Readout noise flips a reported syndrome bit. Circuit noise acts at physical operation locations and can propagate through subsequent gates. These models describe different experiments.

MATHS

Independent bit flips:

$$\mathcal{N}_X = \bigotimes_{j=1}^n \mathcal{N}_{X,j}, \quad \mathcal{N}_{X,j}(\rho) = (1 - p_j)\rho + p_j X \rho X.$$

Single-qubit depolarizing convention:

$$\mathcal{D}_p(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z).$$

$$\Pr(x = 1) = \Pr(z = 1) = 2p/3, \quad \Pr(x = z = 1) = p/3 \text{ / } (2p/3)^2 \text{ in general.}$$

Readout noise:

$$m_{a,t} = s_{a,t} \oplus \eta_{a,t}, \quad \eta_{a,t} \sim \text{Bernoulli}(q).$$

Circuit model: channels $\mathcal{N}_\ell$ at locations ℓ (gates, resets, idles, measurements).

04.06 · Exercise and result

EXPLANATION

Use the two-row parity-check matrix to reproduce the repetition-code syndrome table. Adding the first- and second-qubit syndromes leaves the same result as a flip on the third qubit.

This is the first explicit example of different physical errors with identical evidence. Also notice that a phase flip produces zero syndrome despite changing the logical state.

MATHS

$$H_Z = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad s = H_Z \mathbf{x}.$$

E						
$s(E)$	I	X_1	X_2	X_3	X_1X_2	Z_1
	00	10	11	01	01	00

$$s(X_1X_2) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = s(X_3).$$

Equivalent errors and ambiguous evidence

05.01 · Stabilizers and the normalizer

EXPLANATION

Within the Pauli group, preserving the stabilizer code space is equivalent to commuting with every stabilizer. The Pauli normalizer and centralizer agree here: conjugation by a Pauli only changes another Pauli's sign, and the stabilizer group cannot contain both signs of an element.

An operator in this normalizer can still change the encoded information. The normalizer describes all syndrome-invisible Pauli actions; the stabilizer describes the harmless ones. We explicitly include irrelevant global phases when comparing physical error actions.

MATHS

$$N_{\mathcal{P}_n}(\mathcal{S}) = \{E \in \mathcal{P}_n : ESE^\dagger = \mathcal{S}\},$$

$$C_{\mathcal{P}_n}(\mathcal{S}) = \{E \in \mathcal{P}_n : [E, g] = 0 \forall g \in \mathcal{S}\}.$$

For a stabilizer group with $-I \notin \mathcal{S}$:

$$N_{\mathcal{P}_n}(\mathcal{S}) = C_{\mathcal{P}_n}(\mathcal{S}) =: N(\mathcal{S}).$$

$$E \in N(\mathcal{S}) \iff s(E) = 0 \iff EC = \mathcal{C}.$$

Phase closure: $\tilde{\mathcal{S}} = \{i^q g : q \in \mathbb{Z}_4, g \in \mathcal{S}\}.$

05.02 · Two equivalence relations

EXPLANATION

Errors are equivalent on the code when their difference is a stabilizer, up to phase. A decoder may freely substitute one for the other.

Having the same syndrome is a weaker condition. The difference may be a logical operator rather than a stabilizer. Both errors then lead to the same measured sector while carrying different logical information inside it.

The binary version removes phase bookkeeping and makes the distinction two membership tests: one in the stabilizer subspace, the other in its symplectic orthogonal complement.

MATHS

For $E, F \in \mathcal{P}_n$:

$$E \equiv_c F \iff \exists \theta : FV = e^{i\theta} EV \iff E^\dagger F \in \tilde{\mathcal{S}}.$$

$$s(E) = s(F) \iff E^\dagger F \in N(\mathcal{S}).$$

$$E \equiv_c F \implies s(E) = s(F), \quad s(E) = s(F) \text{ \textcolor{red}{centernot}} \implies E \equiv_c F.$$

Binary form, with $W \subseteq \mathbb{F}_2^{2n}$ the stabilizer subspace:

$$W^\perp = \{e : e^T J w = 0 \forall w \in W\}, \quad W \subseteq W^\perp,$$

$$E(e) \equiv_c E(f) \iff e + f \in W, \quad s(e) = s(f) \iff e + f \in W^\perp.$$

05.03 · Logical Pauli classes

EXPLANATION

Quotienting by stabilizers groups together physical operators that implement the same logical Pauli. There are four phase-free Pauli choices for each encoded qubit.

Although overall phases are omitted from the classes, their commutation relations still matter. The inherited symplectic pairing records which logical representatives anticommute. A logical X and Z for the same encoded qubit have an odd pairing.

MATHS

$$\dim W = n - k, \quad \dim W^\perp = n + k, \quad \dim(W^\perp/W) = 2k.$$

$$\text{Logical Paulis modulo phase} \cong W^\perp/W, \quad |W^\perp/W| = 2^{2k} = 4^k.$$

Representatives $\bar{x}_i, \bar{z}_i \in W^\perp$:

$$\bar{x}_i^T J \bar{x}_j = 0, \quad \bar{z}_i^T J \bar{z}_j = 0, \quad \bar{x}_i^T J \bar{z}_j = \delta_{ij}.$$

$$\bar{X}_i \bar{Z}_j = (-1)^{\delta_{ij}} \bar{Z}_j \bar{X}_i.$$

05.04 · Correction and the residual

EXPLANATION

Recovery is judged by the combination of the actual error and the chosen correction. This combined operator is the residual.

A syndrome-matching correction guarantees that the residual is in the normalizer. Success requires the stronger result that it is a stabilizer. Logical X and Z indicators can be computed by testing the residual against representative logical operators.

A nontrivial logical Pauli can leave one special input eigenstate unchanged. That does not make it a successful quantum memory operation, because the requirement is to preserve every encoded state and its external entanglement.

MATHS

$$R := CE, \quad r = c + e \in \mathbb{F}_2^{2n}, \quad s(R) = s(C) + s(E).$$

Condition	Binary condition	Outcome
$R \in \tilde{\mathcal{S}}$	$r \in W$	success for all code states
$R \in N(\mathcal{S}) \setminus \tilde{\mathcal{S}}$	$r \in W^\perp \setminus W$	logical Pauli remains
$R \notin$	$N(\mathcal{S}) \notin$	$W^\perp$ nonzero syndrome remains

For $k = 1, r \in W^\perp$:

$$\lambda_X = r^T J \bar{z}, \quad \lambda_Z = r^T J \bar{x}, \quad [r] = \lambda_X [\bar{x}] + \lambda_Z [\bar{z}].$$

05.05 · Knill–Laflamme condition

EXPLANATION

The general exact-correction criterion covers arbitrary channels, including coherent errors and degenerate codes. Every pairwise error overlap, restricted to the code space, must be a scalar times the identity there.

The environment formulation explains why. Under this condition, the environment's resulting state does not depend on the logical input. It has learned about the error process without acquiring the encoded information. Degeneracy is allowed: distinct error labels do not need distinct observable outcomes.

MATHS

Let $\mathcal{N}(\rho) = \sum_a E_a \rho E_a^\dagger$ be CPTP.

$$\mathcal{N} \text{ exactly correctable on } \mathcal{C} \iff \exists (c_{ab}) : P E_a^\dagger E_b P = c_{ab} P \quad \forall a, b.$$

$$c = c^\dagger \succeq 0, \quad \sum_a c_{aa} = 1.$$

Stinespring isometry:

$$W_{\mathcal{N}} |\psi_L\rangle = \sum_a E_a |\psi_L\rangle \otimes |a\rangle_E,$$

$$\rho_E = \sum_{a,b} \langle \psi_L | E_b^\dagger E_a | \psi_L \rangle |a\rangle \langle b| = \sum_{a,b} c_{ba} |a\rangle \langle b|.$$

05.06 · Exercise and result

EXPLANATION

When the actual error flips the first two qubits, a decoder that flips the third qubit clears both checks. But it has completed the triple-X logical operator.

This is why passing all checks is weaker evidence than preserving the logical information. The syndrome tells us whether the state has returned to the code space; it does not certify that it returned to the same logical state.

MATHS

Repetition code:

$$\begin{aligned} E &= X_1 X_2, \quad C = X_3, \quad s(E) = s(C) = 01, \\ CE &= X_1 X_2 X_3 = \bar{X}, \quad CE(\alpha|0_L\rangle + \beta|1_L\rangle) = \alpha|1_L\rangle + \beta|0_L\rangle. \\ s(CE) &= 00, \quad CE \notin \tilde{\mathcal{S}}. \end{aligned}$$

Distance and the limits of recovery

06.01 · Weight and distance

EXPLANATION

Weight counts physical qubits on which the error acts nontrivially. A Y has weight one, even though it has both binary components.

Distance is the smallest weight of a syndrome-invisible operator with a nontrivial logical action. A low-weight stabilizer does not reduce distance, since it leaves every codeword unchanged. The three code parameters count physical data qubits, logical qubits, and this minimum logical weight.

MATHS

$$\text{supp}(E) = \{j : P_j \neq I\}, \quad E \sim \bigotimes_{j=1}^n P_j, \quad \text{wt}(E) = |\text{supp}(E)|.$$

$$\text{wt}(e) = \sum_{j=1}^n \mathbf{1}[(x_j, z_j) \neq (0, 0)], \quad \text{wt}(EF) \leq \text{wt}(E) + \text{wt}(F).$$

For $k \geq 1$:

$$d = \min_{L \in N(\mathcal{S}) \setminus \mathcal{S}} \text{wt}(L) = \min_{\ell \in W^\perp \setminus W} \text{wt}(\ell), \quad [[n, k, d]].$$

06.02 · Detection below the distance

EXPLANATION

Every Pauli error below the distance is either harmless on the code or detectable by a syndrome.

“Detectable” does not mean that a harmless stabilizer must set off an alarm.

The zero sandwich in the math pane means the error moves the entire code space into an orthogonal sector. The scalar sandwich means it cannot distinguish or change the logical state, except for an irrelevant overall phase.

MATHS

For Pauli E with $\text{wt}(E) < d$:

$$PEP = \begin{cases} e^{i\theta} P, & E \in \tilde{\mathcal{S}}, \\ 0, & E \notin \tilde{\mathcal{S}} \end{cases} \quad N(\mathcal{S}).$$

For an anticommuting $g \in \mathcal{S}$:

$$PEP = PgEgP = -PEP \implies PEP = 0.$$

06.03 · Guaranteed correction and its proof

EXPLANATION

Correction must distinguish between pairs of possible errors. Two errors each affecting at most a given number of qubits can differ on twice as many qubits. If that combined weight is still below distance, any same-syndrome ambiguity is harmless.

This argument proves existence of a recovery that corrects the whole allowed set, including its linear span. It does not guarantee that an arbitrary implementation or decoder achieves the bound.

To see why the worst-case guarantee ends there, divide a minimum logical operator into two pieces. Those pieces have the same syndrome but differ logically, so one syndrome-only decision cannot repair both.

MATHS

$$\mathcal{E}_t = \{E \in \{I, X, Y, Z\}^{\otimes n} : \text{wt}(E) \leq t\}, \quad 2t < d.$$

$$E, F \in \mathcal{E}_t \implies \text{wt}(E^\dagger F) \leq 2t < d \implies PE^\dagger FP = c_{EF}P.$$

$$t_{\max} = \left\lfloor \frac{d-1}{2} \right\rfloor.$$

d	3	5	7	9
$t_{\max}$	1	2	3	4

Tightness: partition a weight- d logical support into disjoint sets A, B ,

$$L = E_A F_B, \quad |A| = \lfloor d/2 \rfloor, \quad |B| = \lceil d/2 \rceil, \quad s(E_A) = s(F_B), \quad E_A \not\sim F_B. \quad \equiv c$$

06.04 · Restricted noise and asymmetric distance

EXPLANATION

A code can protect one error component much better than the other. For CSS codes, the separate X and Z distances state those strengths explicitly.

The repetition code has a length-three logical X but a weight-one logical Z. Its ability to correct a single bit flip is therefore consistent with having quantum distance one. The general quantum-

distance guarantee covers arbitrary qubit errors, not only the noise type the repetition code handles well.

MATHS

For a CSS code:

$$d_X = \min_{\mathbf{x} \in \ker H_Z \setminus \text{im } H_X^T} |\mathbf{x}|, \quad d_Z = \min_{\mathbf{z} \in \ker H_X \setminus \text{im } H_Z^T} |\mathbf{z}|, \quad d = \min(d_X, d_Z).$$

Repetition code:

$$H_X \text{ has no rows, } H_Z = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad d_X = 3, \quad d_Z = 1, \quad d = 1.$$

$$\{I, X_1, X_2, X_3\} \text{ correctable, } \quad \{I, X_1, Y_1, Z_1, \dots, X_3, Y_3, Z_3\} \text{ not correctable.}$$

06.05 · Known errors and erasures

EXPLANATION

A known Pauli error can always be inverted, even if its weight is large. The difficulty in decoding is uncertainty about which error occurred.

For erasures, the affected positions are known. Comparing two candidate errors then stays within the same erased set instead of combining two unknown supports. This is why every set of fewer than distance-many erased qubits is correctable. Some larger patterns are also correctable when they contain no complete logical support.

MATHS

Known unitary Pauli E :

$$C = E^\dagger \implies CE = I \quad \text{for any weight.}$$

Known erased set $A \subseteq \{1, \dots, n\}$:

$$\mathcal{E}_A = \{E : \text{supp}(E) \subseteq A\}, \quad \text{supp}(E^\dagger F) \subseteq A \quad (E, F \in \mathcal{E}_A).$$

$$|A| < d \implies PE^\dagger FP = c_{EF}P \quad \forall E, F \in \mathcal{E}_A.$$

Exact erasure criterion for stabilizer codes:

$$A \text{ correctable} \iff \nexists L \in N(\mathcal{S}) \setminus \tilde{\mathcal{S}} : \text{supp}(L) \subseteq A.$$

06.06 · Exercise and result

EXPLANATION

Explain the repetition-code exception using its unequal X and Z distances. Then distinguish three claims: a decoder chose the wrong logical class, syndrome information cannot distinguish two possibilities, and information has become inaccessible to the available recovery operation.

These are different meanings sometimes compressed into “unrecoverable.” Weight above the guaranteed correction radius means failure becomes possible. It does not make every such error fail; a stabilizer of that weight is already harmless.

MATHS

$$\text{wt}(E) > t_{\max} \text{ \textcolor{red}{centernot}} \implies \text{decoder failure.}$$

Example: any $g \in \mathcal{S}$ with $\text{wt}(g) > t_{\max}$,

$$E = g, \quad C = I \implies CE = g \sim I \text{ on } \mathcal{C}.$$

$$\text{Known } E \quad / \quad \text{known } s(E) = / \quad \text{known } \textit{operatornamesupp}(E).$$

Building a surface-code patch

07.01 · Cellulation and boundary convention

EXPLANATION

Use the source notes' unrotated convention: data qubits lie on edges, X checks sit at vertices, and Z checks sit on faces. The planar patch has two opposite rough boundaries and two opposite smooth boundaries.

At a rough boundary a Z string may end without a measured vertex defect. At a smooth boundary the corresponding statement holds for an X string on the dual lattice.

The relative-chain formulation makes this boundary rule precise. Cells belonging to the rough boundary are removed from the relevant relative chain groups. It specifies a compatible code construction; simply clipping arbitrary checks from a square drawing need not do so.

MATHS

Let K be a cellulation of a disk and $R \subset \partial K$ the union of two disjoint rough boundary arcs. Remaining boundary arcs: smooth.

$$C_j = C_j(K, R; \mathbb{F}_2), \quad C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0, \quad \partial_1 \partial_2 = 0.$$

$$\text{Data qubits} \leftrightarrow \text{basis of } C_1, \quad H_X = \partial_1, \quad H_Z = \partial_2^T, \quad H_X H_Z^T = 0.$$

Rough: Z endpoints unmeasured, smooth: dual X endpoints unmeasured.

07.02 · Local checks commute

EXPLANATION

Each bulk check involves four nearby data qubits. A vertex and an adjacent face share two edges. Each shared edge contributes a minus sign when X and Z are exchanged, so the two signs cancel.

Boundary supports are modified consistently with the cellulation and boundary type. The matrix identity guarantees commutation throughout the patch. The rank formula counts the logical degrees of freedom left after all independent constraints.

$$A_v = \prod_{e: (\partial_1)_{v,e}=1} X_e, \quad B_f = \prod_{e: (\partial_2)_{e,f}=1} Z_e.$$

$$A_v B_f = (-1)^{\sum_e (\partial_1)_{v,e} (\partial_2)_{e,f}} B_f A_v = B_f A_v.$$

Square-lattice bulk:

$$\text{wt}(A_v) = \text{wt}(B_f) = 4, \quad |\text{supp}(A_v) \cap \text{supp}(B_f)| \in \{0, 2\}.$$

$$n = \dim C_1, \quad k = n - \text{rank } \partial_1 - \text{rank } \partial_2.$$

07.03 · Primal and dual error strings

EXPLANATION

A Z chain follows edges of the primal lattice. Its measured defects occur at vertices with odd incidence. Interior vertices touched by two chain edges contribute twice and cancel.

X chains are drawn on dual edges crossing the data edges. Their defects appear at face checks. The primal and dual pictures describe the two components of the same physical Pauli error; Y contributes to both.

MATHS

$$E = X^{\mathbf{x}} Z^{\mathbf{z}}, \quad s_X = \partial_1 \mathbf{z}, \quad s_Z = \partial_2^T \mathbf{x}.$$

For a primal two-edge path $v_0 \xrightarrow{e_1} v_1 \xrightarrow{e_2} v_2$:

$$\partial_1(e_1 + e_2) = (v_0 + v_1) + (v_1 + v_2) = v_0 + v_2.$$

$$Y_e \sim X_e Z_e \implies s_X(Y_e) = s_X(Z_e), \quad s_Z(Y_e) = s_Z(X_e).$$

07.04 · Logical strings and distance

EXPLANATION

A Z string joining the two distinct rough boundaries can carry the logical Z action. A dual X string joining the two distinct smooth boundaries can carry logical X. Choose representatives that cross once; their physical supports overlap on one qubit, so they anticommute.

Distance is the shortest nontrivial representative, not the length of whichever string was drawn first. Rectangular patches can have unequal distances in the two error sectors.

MATHS

$$\bar{\mathbf{z}} \in \ker H_X \setminus \text{im } H_Z^T, \quad \bar{\mathbf{x}} \in \ker H_Z \setminus \text{im } H_X^T.$$

$$\bar{Z} = Z^{\bar{z}}, \quad \bar{X} = X^{\bar{x}}, \quad \bar{\mathbf{x}}^T \bar{\mathbf{z}} = 1 \implies \bar{X} \bar{Z} = -\bar{Z} \bar{X}.$$

$$d_Z = \min_{\mathbf{z} \in \ker H_X \setminus \text{im } H_Z^T} |\mathbf{z}|, \quad d_X = \min_{\mathbf{x} \in \ker H_Z \setminus \text{im } H_X^T} |\mathbf{x}|.$$

For a disk with exactly two rough arcs and two smooth arcs:

$$k = 1, \quad d = \min(d_X, d_Z).$$

07.05 · Rotated patches and measurement hardware

EXPLANATION

The rotated layout packs the same kind of topological protection into a different geometry. In the usual odd-distance square patch, there is one logical qubit, a square array of data qubits, four-body interior checks, and two-body boundary checks.

The code parameters count data qubits. Measurement ancillas are additional hardware. The total shown assumes one dedicated ancilla per check; it is a hardware convention rather than a universal total including routing or other support qubits.

Do not carry edge coordinates or boundary supports from an unrotated drawing straight into a rotated implementation. The algebraic check matrices must match the chosen geometry.

MATHS

Standard odd- d rotated square family, $d \geq 3$:

$$[[n, k, d]] = [[d^2, 1, d]], \quad r_X = r_Z = \frac{d^2 - 1}{2}, \quad r_X + r_Z = d^2 - 1.$$

$$\text{wt}(g) = \begin{cases} 4, & \text{bulk check,} \\ 2, & \text{boundary check.} \end{cases}$$

One dedicated ancilla per check:

$$n_{\text{data}} = d^2, \quad n_{\text{anc}} = d^2 - 1, \quad n_{\text{data}} + n_{\text{anc}} = 2d^2 - 1.$$

$$d = 3 : \quad 9 \text{ data} + 8 \text{ ancillas} = 17 \text{ qubits.}$$

07.06 · Exercise and result

EXPLANATION

Track a Z chain as a second edge is added. The shared interior vertex loses its defect because the two anticommutations cancel.

If one endpoint lies on the rough boundary, only the other endpoint appears in the measured syndrome. A single visible defect is therefore a legitimate pattern. It is not evidence that a second defect was accidentally omitted.

MATHS

$$Z_{e_1} : s_X = v_0 + v_1, \quad Z_{e_1} Z_{e_2} : s_X = v_0 + v_2.$$

If $v_0 \in R$:

$$[v_0] = 0 \text{ in } C_0(K, R), \quad \partial_1(e_1 + e_2) = [v_2].$$

Drawing boundaries and understanding topology

08.01 · Relative chains

EXPLANATION

A chain is a set of edges with binary coefficients. Addition toggles membership: using an edge twice cancels it. Its boundary records vertices touched by an odd number of those edges.

Relative chains treat rough-boundary endpoints as zero in the boundary space. This allows an apparently open string to have zero measured boundary.

Drawing the boundary of an error means marking these checked endpoints. It does not mean outlining a region around the qubits. The three-edge example is an exact symbolic version of the original endpoint diagram.

MATHS

$$C_j(K, R; \mathbb{F}_2) = C_j(K; \mathbb{F}_2) / C_j(R; \mathbb{F}_2).$$

$$\partial_j^{\text{rel}}[a] = [\partial_j a], \quad \partial_1^{\text{rel}} \partial_2^{\text{rel}} = 0.$$

Suppressing the superscript:

$$e = \sum_q e_q q \in C_1, \quad e_q \in \mathbb{F}_2, \quad s = \partial_1 e, \quad e + e = 0.$$

Three-edge chain:

$$\partial_1(e_{01} + e_{12} + e_{23}) = (v_0 + v_1) + (v_1 + v_2) + (v_2 + v_3) = v_0 + v_3.$$

08.02 · Stabilizer deformations

EXPLANATION

Multiplying an error by a face stabilizer toggles every edge around that face. A path using one edge can be replaced by the other three sides.

The syndrome is unchanged because a face boundary has no remaining boundary. More strongly, the new error has exactly the same action on the code. The visual path deformation is therefore a concrete stabilizer multiplication.

MATHS

$$b = \partial_2 f, \quad Z^b \in \mathcal{S}, \quad e' = e + \partial_2 f.$$

$$\partial_1 e' = \partial_1 e + \partial_1 \partial_2 f = \partial_1 e.$$

Square face with $\partial_2 f = a + b + c + d$:

$$e = a \implies e' = a + (a + b + c + d) = b + c + d.$$

$$Z^{e'} V = Z^e Z^{\partial_2 f} V = Z^e V.$$

08.03 · Residual cycles and logical classes

EXPLANATION

A correction with matching endpoints makes the residual a cycle in the relative sense. The remaining question is whether that cycle is a sum of face boundaries.

If it is, the residual is a stabilizer and recovery succeeds in this error sector. If it is not, it represents a nontrivial homology class and changes the logical information.

This is the topology behind the normalizer-versus-stabilizer distinction. The chain language specializes the same algebra to a geometric code.

MATHS

$$\partial_1 c = \partial_1 e = s \implies r = e + c \in Z_1 := \ker \partial_1.$$

$$B_1 := \text{im } \partial_2 \subseteq Z_1, \quad H_1(K, R; \mathbb{F}_2) = Z_1 / B_1.$$

$$\text{Z-sector success} \iff [r] = 0 \iff r \in B_1, \quad \text{Z logical failure} \iff [r] \neq 0.$$

$$\dim H_1 = \dim C_1 - \text{rank } \partial_1 - \text{rank } \partial_2 = k.$$

08.04 · Planar and closed surfaces

EXPLANATION

A disk has no ordinary noncontractible loops, yet a planar patch can encode a qubit. The missing ingredient is the relative boundary condition. A string connecting the two separate rough arcs is closed for the measured boundary operator and cannot be removed by face deformations.

The short homology calculation counts one independent such class. A torus instead has two independent ordinary loop directions and encodes two qubits.

For the disk with no holes considered here, a path whose endpoints lie on the same rough arc is relatively trivial. Boundary type and connectivity matter; arbitrary boundary-to-boundary paths are not all logical operators.

MATHS

Disk $K, R = R_1 \sqcup R_2$ two contractible disjoint boundary arcs:

$$H_1(K; \mathbb{F}_2) = 0, \quad H_1(K, R; \mathbb{F}_2) \cong \ker(H_0(R) \rightarrow H_0(K)).$$

$$\mathbb{F}_2^2 \rightarrow \mathbb{F}_2 : (a, b) \mapsto a + b, \quad H_1(K, R; \mathbb{F}_2) \cong \mathbb{F}_2.$$

For a chain γ joining R_1 to R_2 :

$$\partial_1^{\text{rel}} \gamma = 0, \quad [\gamma] \neq 0. \quad =$$

Torus:

$$H_1(T^2; \mathbb{F}_2) \cong \mathbb{F}_2^2, \quad k = 2.$$

08.05 · Dual sector and intersection pairing

EXPLANATION

The X sector is naturally the cohomological partner of the Z-chain description. Geometrically it can be drawn as dual paths terminating at smooth boundaries.

Their intersection parity determines commutation. Changing either representative by a stabilizer leaves that parity unchanged, so the pairing depends on logical classes rather than drawing details.

For a general Pauli error, recovery must be trivial in both sectors. Fixing only the Z-chain homology is insufficient when an X component remains.

MATHS

Using cell bases to identify cochains with binary column vectors:

$$H^1(K, R; \mathbb{F}_2) = \ker \partial_2^T / \text{im } \partial_1^T.$$

$$r_Z \in \ker \partial_1, \quad r_X \in \ker \partial_2^T, \quad \langle [r_X], [r_Z] \rangle = r_X^T r_Z \pmod{2}.$$

Representative invariance:

$$(r_X + \partial_1^T a)^T r_Z = r_X^T r_Z, \quad r_X^T (r_Z + \partial_2 b) = r_X^T r_Z.$$

$$X^{r_X} Z^{r_Z} = (-1)^{r_X^T r_Z} Z^{r_Z} X^{r_X}.$$

08.06 · Exercise and result

EXPLANATION

Draw a path and deform it around a face to make the first correction candidate. Then toggle a full rough-to-rough logical path to make a second candidate.

Both corrections match the measured endpoints. The first differs from the original by a stabilizer, while the second differs by that stabilizer plus a logical string. Matching endpoints is necessary for clearing the syndrome, but the residual class determines whether the repair succeeds.

MATHS

Choose $e, c \in C_1$ and a nontrivial relative cycle ℓ :

$$c = e + \partial_2 f, \quad c' = c + \ell, \quad \partial_1 \ell = 0, \quad [\ell] \neq 0. \quad =$$

$$\partial_1 c = \partial_1 c' = \partial_1 e,$$

$$[e + c] = [\partial_2 f] = 0, \quad [e + c'] = [\partial_2 f + \ell] = [\ell] \neq 0. \quad =$$

Decoding from imperfect evidence

09.01 · Inference from a syndrome

EXPLANATION

The decoder sees a syndrome and combines it with a noise model. Maximum a posteriori error inference chooses the most probable compatible physical error.

The correction is the inverse of that estimate. It can succeed even when the estimate is not the actual error, provided their difference is a stabilizer. All probability formulas in this lesson assume a stochastic Pauli model; they do not directly describe coherent superpositions of physical errors.

MATHS

Assume a stochastic Pauli distribution $\pi(e)$ and ideal syndrome $s = HJe$.

$$\Pr(e \mid s) = \frac{\pi(e) \mathbf{1}[HJe = s]}{\sum_{f: HJf = s} \pi(f)}.$$

$$\hat{e}_{\text{MAP}} \in \arg \max_{e: HJe = s} \pi(e), \quad C = E(\hat{e}_{\text{MAP}})^\dagger.$$

$$\hat{e}_{\text{MAP}} / e \text{ \textcolor{red}{centernot} } \implies \text{logical failure}.$$

09.02 · Logical-class inference

EXPLANATION

All errors compatible with a syndrome split into logical classes. Errors within a class differ by stabilizers and admit the same successful recovery action.

To optimize the probability of preserving arbitrary logical information, sum the probabilities of every physical error in each class and choose the largest total. This can differ from choosing the single most probable error: many moderately likely errors may together outweigh one especially likely representative.

The choice of reference error changes the labels assigned to the classes, not the physical recovery problem.

MATHS

Fix e_s with $HJe_s = s$. Choose representatives ℓ of $W^\perp / W$.

$$\{e : HJe = s\} = \bigsqcup_{[\ell] \in W^\perp / W} (e_s + \ell + W).$$

$$Z_{[\ell]}(s) = \sum_{w \in W} \pi(e_s + \ell + w), \quad \Pr([\ell] \mid s) = \frac{Z_{[\ell]}(s)}{\sum_{[\ell']} Z_{[\ell']}(s)}.$$

$$[\hat{\ell}] \in \arg \max_{[\ell]} Z_{[\ell]}(s), \quad C = E(e_s + \hat{\ell})^\dagger.$$

$$P_{\text{succ}}^{\text{opt}}(s) = \max_{[\ell]} \Pr([\ell] \mid s).$$

09.03 · Independent faults become additive weights

EXPLANATION

Independent binary faults turn likelihood into a sum of edge costs. A rare fault gets a large positive cost; a more likely fault gets a smaller one. Uniform probabilities below one half reduce the objective to counting faults.

This derivation explains minimum weight. It also states its assumptions. Correlated faults generally do not factor into these independent costs, and optimizing a most likely fault configuration need not optimize the summed probability of a logical class.

MATHS

For binary fault variables $u_j \in \{0, 1\}$, independently sampled with $0 < p_j < 1$:

$$\Pr(u) = \prod_j p_j^{u_j} (1 - p_j)^{1-u_j},$$

$$-\log \Pr(u) = -\sum_j \log(1 - p_j) + \sum_j u_j w_j, \quad w_j = \log \frac{1 - p_j}{p_j}.$$

For detector-incidence matrix B :

$$\hat{u} \in \arg \min_{u: Bu=s} \sum_j w_j u_j.$$

$$p_j = p < 1/2 \implies w_j = w > 0 \implies \hat{u} \in \arg \min_{u: Bu=s} |u|.$$

09.04 · Graphlike models and matching

EXPLANATION

For a graphlike model, a fault has two measured endpoints, one endpoint and a permitted boundary, or no measured endpoints. The decoding task asks for an edge set with the observed odd-degree vertices.

A boundary-aware matching construction pairs defects with other defects or with legal boundaries. Shortest-path distances provide the pairing costs. Expanding the chosen pairings and adding paths modulo two produces a candidate fault chain.

Boundary nodes need the proper matching construction, such as virtual copies; one ordinary boundary vertex cannot simply be forced to accept only one defect. Fault labels must also track effects on logical observables. A zero-endpoint fault can still be logically harmful even though the detector record cannot locate it.

MATHS

Assume $0 < p_j < 1/2$ and each modeled fault has at most two measured endpoints:

$$|B_{:,j}| \in \{0, 1, 2\}, \quad T = \{v : s_v = 1\}.$$

For measured vertices V_m and permitted boundary vertices V_b :

$$\text{odd}_{V_m}(F) = T, \quad F_* \in \arg \min_{F: \text{odd}_{V_m}(F)=T} \sum_{e \in F} w_e.$$

Shortest-path metric:

$$d_w(a, b) = \min_{\gamma: a \rightsquigarrow b} \sum_{e \in \gamma} w_e, \quad d_w(a, V_b) = \min_{b \in V_b} d_w(a, b).$$

Boundary-aware matching reduction:

$$M_* \in \arg \min_{M \text{ feasible}} \sum_{(a,b) \in M} d_w(a, b), \quad \hat{u} = \bigoplus_{(a,b) \in M_*} \mathbf{1}_{\gamma_{ab}}.$$

Fault-label map L :

$$\hat{\lambda} = L\hat{u}, \quad \lambda_{\text{residual}} = L(u + \hat{u}).$$

09.05 · Model limitations and correlated components

EXPLANATION

Decoding the two CSS components independently is convenient, but Y errors correlate them. A decoder using those correlations can make a different inference.

Some circuit faults create more than two detector events. Representing such a fault as unrelated graph edges can change its probability structure. Correlation-aware matching methods or other decoding approaches are needed when the simple independent graph model is inadequate.

Even under the correct model, an optimal decoder can fail on a particular sample. It chooses from incomplete evidence rather than observing the hidden fault directly.

Depolarizing convention:

$$\Pr(x_j, z_j) = \begin{cases} 1 - p, & (x_j, z_j) = (0, 0), \\ p/3, & (x_j, z_j) \in \{(1, 0), (0, 1), (1, 1)\}. \end{cases}$$

$$\Pr(\mathbf{x}, \mathbf{z}) / \Pr(\mathbf{x}) \Pr(\mathbf{z}) \quad \text{in general.}$$

$$\left(\arg \max_{\mathbf{x}} \Pr(\mathbf{x} \mid s_Z), \arg \max_{\mathbf{z}} \Pr(\mathbf{z} \mid s_X) \right) / \arg \max_{\mathbf{x}, \mathbf{z}} \Pr(\mathbf{x}, \mathbf{z} \mid s_X, s_Z) \quad \text{in general.} \quad =$$

$$|B_{:,j}| > 2 \implies \text{fault } j \text{ is not a single ordinary graph edge.}$$

09.06 · Pauli frame

EXPLANATION

A Pauli frame records the correction in classical software instead of applying it immediately to the hardware. To remain equivalent, later gates and measurement outcomes must be interpreted using the frame.

Clifford gates map Paulis to Paulis, so their frame update is especially simple. For a Pauli measurement, an anticommuting frame flips the interpretation of its sign. A deferred correction remains part of the computation.

MATHS

Convention: stored frame F maps the tracked physical state to the corrected reference,

$$\rho_{\text{corr}} = F \rho_{\text{phys}} F^\dagger.$$

Clifford U :

$$F' = U F U^\dagger \implies F'(U \rho_{\text{phys}} U^\dagger) F'^\dagger = U \rho_{\text{corr}} U^\dagger.$$

Pauli measurement M , outcome $m_{\text{phys}} \in \{\pm 1\}$:

$$F M = (-1)^\omega M F \implies m_{\text{corr}} = (-1)^\omega m_{\text{phys}}.$$

09.07 · Exercise and result

EXPLANATION

The third-qubit flip is more likely than the first-two-qubit flip when the physical bit-flip probability is below one half. It is therefore the sensible correction for this syndrome.

Nevertheless, the two-flip event sometimes happens. The decoder then applies the wrong logical class. The final formula counts all two- and three-flip patterns, which are precisely the failures of this restricted repetition-code decoder.

MATHS

Repetition code, independent bit flips, $0 < p < 1/2$, syndrome 01:

$$\Pr(X_3) = p(1-p)^2, \quad \Pr(X_1X_2) = p^2(1-p),$$

$$\frac{\Pr(X_3)}{\Pr(X_1X_2)} = \frac{1-p}{p} > 1.$$

$$\Pr(X_3 \mid 01) = 1-p, \quad \Pr(X_1X_2 \mid 01) = p.$$

For the minimum-weight bit-flip decoder:

$$P_{L,X} = \binom{3}{2} p^2(1-p) + p^3 = 3p^2 - 2p^3.$$

When the checks themselves are noisy

10.01 · Stationary repeated measurements

EXPLANATION

With faulty readout, a single negative check result could come from the data or from the measurement. Repeating fixed checks lets us compare neighboring rounds.

The detector record measures changes, not the current error itself. The equation separates new data faults from measurement errors in the current and previous rounds. A pre-existing data error does not keep producing fresh events forever.

The first round needs a known reference or a specified initialization rule. Arbitrarily setting it to zero would invent evidence if the initial syndrome were unknown.

MATHS

Assumptions: fixed checks, Pauli data faults, binary readout noise, discrete rounds.

$$e_t = e_{t-1} + f_t, \quad s_t = HJ e_t, \quad m_t = s_t + \eta_t \quad (\mathbb{F}_2).$$

$$d_t := m_t + m_{t-1} = HJ f_t + \eta_t + \eta_{t-1}.$$

For an initialized known syndrome:

$$m_0 = s_0 \quad \text{if the initial reference is noiseless and known.}$$

10.02 · Spatial and temporal signatures

EXPLANATION

A persistent data error changes the true syndrome when it arrives. Adjacent-round differences register its onset; later identical outcomes cancel.

One bad measurement creates a false change when it appears and another when the next correct readout returns. These neighboring-time events suggest a temporal edge. A data fault with two spatial endpoints suggests a spatial edge in the simplified graph.

A fault near the start or end may have only one visible endpoint, depending on the experiment's time boundary.

MATHS

Single data fault $f_\tau = f, f_t = 0$ for $t \neq \tau, \eta_t = 0$:

$$d_\tau = HJf, \quad d_t = 0 \quad (t \neq \tau).$$

Single readout fault at check a , round τ :

$$\begin{aligned} \eta_\tau &= \mathbf{e}_a, \quad \eta_t = 0 \quad (t \neq \tau), \quad f_t = 0, \\ d_\tau &= \mathbf{e}_a, \quad d_{\tau+1} = \mathbf{e}_a, \quad d_t = 0 \quad (t \notin \{\tau, \tau+1\}). \end{aligned}$$

10.03 · General detectors and time boundaries

EXPLANATION

A general detector is a parity of recorded measurements that has a known result in the ideal experiment. Adjacent rounds of one fixed check are only one example.

Initialization, final data measurements, changed checks, and logical gate operations determine which parities are valid. Some final readouts reveal one logical observable while leaving its conjugate unmeasured.

In circuit decoding, the output can be a predicted logical-observable flip rather than an explicit physical correction path. Detector incidence and observable labels must come from the same circuit model.

MATHS

For measurement-record bits m_j and a deterministic ideal parity b_D :

$$d_D = b_D \oplus \bigoplus_{j \in D} m_j, \quad \bigoplus_{j \in D} m_j^{\text{ideal}} = b_D.$$

$$d = B_{\text{det}} u, \quad \lambda = L_{\text{obs}} u \quad \text{for a specified binary fault model.}$$

$$\hat{u} = \mathcal{D}(d), \quad \lambda_{\text{residual}} = L_{\text{obs}}(u + \hat{u}).$$

10.04 · Three levels of noise

EXPLANATION

Code-capacity experiments isolate data-error decoding by assuming perfect checks.

Phenomenological models add simplified readout errors without simulating every syndrome-extraction gate.

Circuit-level models place faults at the actual physical operations. Their detector patterns include propagation and timing effects. An error probability in one model generally does not describe the same physical quantity as the same numerical value in another, so their thresholds need separate labels.

MATHS

Model	Fault variables	Syndrome information
Code capacity	data Paulis	perfect
Phenomenological	data Paulis, readout bits	noisy repeated checks
Circuit level	gate, idle, reset, measurement faults	circuit-derived detectors

$$\mathcal{N}_{\text{circuit}} = \mathcal{N}_L \circ \mathcal{U}_L \circ \cdots \circ \mathcal{N}_1 \circ \mathcal{U}_1 \quad (\text{with instruments for measurements}).$$

10.05 · CNOT propagation and hook errors

EXPLANATION

CNOT spreads an X on its control to its target, and a Z on its target back to its control. The complementary Pauli directions do not spread in the same way.

A syndrome ancilla interacts with several data qubits, so one ancilla fault can produce a correlated data pattern. Such hook errors can make a logical direction easier to damage than a count of data-qubit distance alone suggests.

The order of check interactions determines the geometry of these correlated patterns. A fault-tolerant schedule must analyze them rather than assuming the ideal stabilizer weight tells the whole story.

MATHS

Let $U = \text{CNOT}_{c \rightarrow t}$:

$$UX_cU^\dagger = X_cX_t, \quad UZ_cU^\dagger = Z_c,$$

$$UX_tU^\dagger = X_t, \quad UZ_tU^\dagger = Z_cZ_t.$$

Ancilla-control X_a fault before remaining targets Q_{rem} :

$$X_a \mapsto X_a \prod_{j \in Q_{\text{rem}}} X_j.$$

Ancilla-target Z_a fault before remaining controls Q_{rem} :

$$Z_a \mapsto Z_a \prod_{j \in Q_{\text{rem}}} Z_j.$$

10.06 · Space, time, and fault distance

EXPLANATION

Spatial code distance is not the entire protection of a noisy measurement experiment. A damaging fault pattern may exploit time boundaries or repeated faulty readout as well as a path across the patch.

The fault-distance definition counts the fewest modeled fault mechanisms that can change the tracked logical observable while producing no detectors. It depends on the circuit and the allowed faults.

Many surface-code protocols use a number of rounds proportional to spatial distance so temporal protection scales too. The required constant and boundary handling belong to the specific memory or gate protocol.

MATHS

For a fixed binary circuit-fault model:

$$d_{\text{fault}} := \min_{u: B_{\text{det}} u = 0, L_{\text{obs}} u \neq 0} |u|.$$

$$d_{\text{fault}} = d_{\text{fault}}(d, T, \text{schedule, time boundaries, fault model}).$$

Common memory scaling choice:

$$T(d) = \lceil \kappa d \rceil, \quad \kappa > 0.$$

10.07 · Exercise and result

EXPLANATION

Insert one flipped readout into five ideal zero outcomes. The detector fires when that wrong value appears and again when it disappears.

The two displayed result lists differ only in whether the known initial reference is included. Stating that convention avoids an apparent off-by-one disagreement in a simulator.

MATHS

$$(m_1, m_2, m_3, m_4, m_5) = (0, 0, 1, 0, 0), \quad m_0 = 0.$$

$$(d_1, d_2, d_3, d_4, d_5) = (0, 0, 1, 1, 0).$$

$$(m_2 \oplus m_1, m_3 \oplus m_2, m_4 \oplus m_3, m_5 \oplus m_4) = (0, 1, 1, 0).$$

Thresholds and scaling protection

11.01 · Specify the logical failure event

EXPLANATION

Choose what counts as failure before collecting data. You might count residual logical X , residual logical Z , either component, or disagreement of one final observable.

These quantities are related but different. A logical Y contributes to both component events and only once to the either-component event. Reading logical Z detects the X component of a residual, while a pure Z residual can remain invisible to that readout.

For a full quantum-memory claim, preserving one prepared logical eigenstate is insufficient. The experiment must test the relevant complementary information or use a simulation criterion that tracks the full logical class.

MATHS

Fix family $\{\mathcal{C}_d\}$, noise $\mathcal{N}_p$, protocol with T rounds, and decoder $\mathcal{D}$.

$$(\lambda_X, \lambda_Z) \in \mathbb{F}_2^2,$$

$$P_{L,X} = \Pr(\lambda_X = 1), \quad P_{L,Z} = \Pr(\lambda_Z = 1),$$

$$P_{L,\text{any}} = \Pr((\lambda_X, \lambda_Z) \neq (0, 0)) = P_{L,X} + P_{L,Z} - \Pr(\lambda_X = \lambda_Z = 1).$$

Logical $\bar{Z}$ readout detects λ_X ; logical $\bar{X}$ readout detects λ_Z .

$$P_L = P_L(p, d, T; \mathcal{N}, \mathcal{D}, \text{protocol, failure event}).$$

11.02 · A threshold for a specified scaling experiment

EXPLANATION

A threshold belongs to a code family, noise model, decoder, and protocol. The definition here asks whether the logical failure probability tends to zero as distance grows under a specified rule for experiment duration.

The time rule is essential. Increasing distance while allowing arbitrarily long storage can change the conclusion. Exponential suppression with distance can overcome polynomial growth in storage duration, as the displayed bound illustrates.

This definition states an asymptotic property. It is not a universal numerical constant attached to the words “surface code.”

MATHS

Fix a time-scaling rule $T = T(d)$ and write $P_L^*(p, d) = P_L(p, d, T(d))$.

$$p_{\text{th}} := \sup \left\{ p_0 \in [0, 1] : \forall p \in [0, p_0), \lim_{d \rightarrow \infty} P_L^*(p, d) = 0 \right\}.$$

A possible below-threshold bound:

$$P_L^*(p, d) \leq A(p)e^{-\alpha(p)d}, \quad \alpha(p) > 0.$$

If $P_L(p, d, T) \leq TA(p)e^{-\alpha(p)d}$ and $T(d) = O(d^a)$:

$$\lim_{d \rightarrow \infty} P_L(p, d, T(d)) = 0.$$

11.03 · Finite-size crossings

EXPLANATION

Curves for several distances often cross near a threshold because the leading scaling variable vanishes there. Finite-size corrections shift those crossings.

The scaling equation is an ansatz to justify and fit for a particular model, not a theorem for every decoder or noise process. Consistent protocol aspect ratios matter, especially the relationship between time and spatial distance.

A crossing between two small patches is useful evidence, but it does not by itself give a precise asymptotic threshold. Check sensitivity to the distances included, model assumptions, and statistical uncertainty.

MATHS

Conditional finite-size scaling ansatz, fixed protocol aspect ratios:

$$P_L^*(p, d) \approx F((p - p_{\text{th}})d^{1/\nu}) + d^{-\omega}G((p - p_{\text{th}})d^{1/\nu}), \quad \nu, \omega > 0.$$

$$P_L^*(p_{\text{th}}, d) \approx F(0) + d^{-\omega}G(0).$$

$$P_L^*(p, d_1) = P_L^*(p, d_2) \quad \text{\textcolor{red}{\textbackslash centernot}} \implies \quad p = p_{\text{th}}.$$

11.04 · Low-noise behavior

EXPLANATION

At low noise, the smallest failing fault patterns dominate. If a decoder corrects every pattern up to the guaranteed radius and fails on some patterns one fault larger, that next weight sets the leading power of the error probability.

The familiar distance-based power law expresses this intuition, but its prefactor, duration dependence, and useful range depend on the experiment. Circuit fault weight can differ from data-error weight, and correlations can alter the expansion.

Use the heuristic to interpret measured behavior, not to manufacture a threshold estimate without simulations or a justified model.

MATHS

For a fixed finite stochastic fault model with finitely many locations,

$$P_L(p) = \sum_{u:\text{failure}} p^{|u|} (1-p)^{M-|u|} \quad (\text{equal independent binary probabilities}).$$

If $w_{\min}$ is the minimum failing fault weight:

$$P_L(p) = a_{w_{\min}} p^{w_{\min}} + O(p^{w_{\min}+1}) \quad (p \rightarrow 0).$$

If all errors of weight $\leq t$ are corrected and some weight- $t+1$ errors fail:

$$w_{\min} = t + 1, \quad t = (d-1)/2 \quad (d \text{ odd}).$$

Common heuristic:

$$P_L \approx A \left(\frac{p}{p_{\text{th}}} \right)^{(d+1)/2}.$$

11.05 · Sampling and uncertainty

EXPLANATION

Count failures over independent repetitions of the same experiment. The estimate is the observed fraction, and the uncertainty is binomial.

A Wilson interval gives a practical approximate two-sided interval. For zero observed failures, the separate exact one-sided formula gives an upper confidence bound. These are different interval conventions and should be labeled accordingly.

Zero observed failures means the experiment has not resolved a nonzero rate, not that the true rate is zero. Rare-event estimates need enough samples to make their uncertainty meaningful; reusing correlated trials would invalidate the simple binomial model.

MATHS

Independent identical trials:

$$F \sim \text{Binomial}(N, P_L), \quad \hat{P}_L = f/N, \quad \text{Var}(\hat{P}_L) = P_L(1 - P_L)/N.$$

Wilson interval, normal quantile z :

$$c = \frac{\hat{P}_L + z^2/(2N)}{1 + z^2/N}, \quad h = \frac{z\sqrt{\hat{P}_L(1 - \hat{P}_L)/N + z^2/(4N^2)}}{1 + z^2/N}, \quad [c - h, c + h].$$

For $f = 0$, exact one-sided confidence $1 - \alpha$ upper bound:

$$(1 - P_U)^N = \alpha \implies P_U = 1 - \alpha^{1/N} \approx -\log(\alpha)/N.$$

$$\alpha = 0.05 : \quad P_U \approx 2.9957/N.$$

11.06 · Per-round and per-experiment quantities

EXPLANATION

A final parity error counts an odd number of logical flips. The event “at least one logical flip occurred” also counts even numbers of flips, so these probabilities differ.

Dividing a per-experiment failure fraction by the number of rounds is only a low-error approximation under suitable assumptions. Actual decoder outputs can be temporally correlated and need not follow either independent-round model exactly.

Record the circuit, duration, noise convention, decoder, random seeds, shot counts, failure definition, and uncertainty method. Those details determine what a plotted number means and whether another experiment can reproduce it.

MATHS

Independent flips of one logical parity, probability q each round:

$$\Pr(\text{odd number of flips in } T) = \frac{1 - (1 - 2q)^T}{2}.$$

$$\Pr(\text{at least one flip in } T) = 1 - (1 - q)^T.$$

$$qT \ll 1 \implies \frac{1 - (1 - 2q)^T}{2} \approx Tq, \quad 1 - (1 - q)^T \approx Tq.$$

Reproducibility record:

(code/circuit, d , T , noise convention, p , decoder/version, seeds, N , f , failure event, interval method).

11.07 · Exercise and result

EXPLANATION

Comparing distance three for three rounds with distance five for one hundred rounds confounds spatial protection with exposure time. Even equal per-round logical error probabilities would give very different per-experiment failure fractions.

Use a common duration, a justified common rate model, or a declared scaling rule such as a fixed time-to-distance ratio. If the goal is threshold scaling, state that rule and interpret the curves within it.

MATHS

$$q_3 = q_5 = q, \quad P_3 = \frac{1 - (1 - 2q)^3}{2}, \quad P_5 = \frac{1 - (1 - 2q)^{100}}{2}.$$

$$q \ll 1/100 \implies P_3 \approx 3q, \quad P_5 \approx 100q, \quad P_5/P_3 \approx 100/3.$$

Computing while remaining protected

12.01 · Logical implementation

EXPLANATION

An ideal logical implementation must produce the intended operation on every encoded input. If the protocol changes the geometry, the output encoding may differ from the input encoding.

Preserving the code space alone is a weaker claim: many different logical gates preserve the same space. The equation specifying the logical action identifies which gate was implemented.

A measurement-based gate has several outcome branches. After their prescribed corrections, all branches must realize the same logical operation, with branch amplitudes independent of the unknown logical input.

MATHS

Encodings V_{in} , V_{out} and ideal logical unitary U_L :

$$U_{\text{phys}} V_{\text{in}} = V_{\text{out}} U_L.$$

For the same input and output code:

$$U_{\text{phys}} P U_{\text{phys}}^\dagger = P, \quad V^\dagger U_{\text{phys}} V = U_L.$$

Measurement-based implementation, corrected branch m :

$$F_m K_m V_{\text{in}} = a_m V_{\text{out}} U_L, \quad \sum_m |a_m|^2 = 1.$$

12.02 · Fault tolerance is an additional condition

EXPLANATION

Fault tolerance concerns what happens when faults occur during the gate, possibly in addition to errors already present. A noiseless gate identity cannot answer that question.

The mathematical pane gives one sufficient test for a fully specified gadget: allowed input errors and allowed fault branches must still yield the intended logical operation after the prescribed frame update and ideal output recovery. The scalar allows an unnormalized measurement or fault branch.

Actual fault-tolerance definitions vary with the gadget and noise model. State which input errors, fault locations, correlated fault mechanisms, and output recovery are covered. A spatial string representing a gate does not supply this circuit-level analysis.

MATHS

Example of a sufficient, explicitly scoped gadget criterion: let $\mathcal{E}_s$ be allowed input errors and $K_{F,m}$ a branch with a specified fault pattern F .

$$\forall E \in \mathcal{E}_s, \forall F : |F| \leq r, \quad s + r \leq t,$$

$$\mathcal{D}_{\text{out}} \circ \mathcal{F}_m \circ \mathcal{K}_{F,m} \circ \mathcal{E} \circ \mathcal{V}_{\text{in}} = q_{E,F,m} \mathcal{U}_L,$$

$$\mathcal{K}_{F,m}(\rho) = K_{F,m} \rho K_{F,m}^\dagger, \quad \mathcal{F}_m(\rho) = F_m \rho F_m^\dagger, \quad q_{E,F,m} \geq 0.$$

Here $\mathcal{D}_{\text{out}}$ is ideal output recovery followed by decoding; admissible fault branches and normalization are part of the specified fault model.

12.03 · Logical Paulis and frame updates

EXPLANATION

A representative logical string implements a logical Pauli. It has the same physical form as an undetected logical error; the difference is whether that transformation is intended and included in the computation's record.

A logical Pauli can often be implemented by updating the frame rather than pulsing every data qubit. That changes how subsequent operations and measurements are interpreted. The frame remains a mathematical description of the intended corrected state.

MATHS

$$\bar{X}V = VX, \quad \bar{Z}V = VZ, \quad \bar{X}\bar{Z} = -\bar{Z}\bar{X}.$$

$$V|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle,$$

$$\bar{X}V|\psi\rangle = \alpha|1_L\rangle + \beta|0_L\rangle, \quad \bar{Z}V|\psi\rangle = \alpha|0_L\rangle - \beta|1_L\rangle.$$

Intended Pauli P_L , frame convention $\rho_{\text{ref}} = F\rho_{\text{phys}}F^\dagger$:

$$F_{\text{new}} = \bar{P}_L F \implies \rho_{\text{ref,new}} = \bar{P}_L \rho_{\text{ref}} \bar{P}_L^\dagger.$$

12.04 · Hadamard and changed geometry

EXPLANATION

Applying Hadamard to every data qubit exchanges X and Z checks on the same supports. The resulting state generally belongs to a code with exchanged check types and boundary roles.

To call the complete planar operation a logical Hadamard in the original architecture, specify how the new geometry is interpreted or restored. Depending on the protocol, that may involve relabeling, movement, or code deformation. It is not supplied by the symbol for transversal Hadamard alone.

A pre-existing Z frame becomes an X frame after logical Hadamard. Applying Hadamard to just one data qubit does not implement this logical transformation.

MATHS

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad HXH = Z, \quad HZH = X.$$

$$H^{\otimes n} X^{\mathbf{a}} H^{\otimes n} = Z^{\mathbf{a}}, \quad H^{\otimes n} Z^{\mathbf{b}} H^{\otimes n} = X^{\mathbf{b}}.$$

$$(H_X, H_Z) \mapsto (H_Z, H_X), \quad \mathcal{C} \mapsto \mathcal{C}'.$$

If a specified geometry-restoring operation G exists with the required logical action:

$$GH^{\otimes n}V = VH_L.$$

Frame conjugation:

$$H_L Z_L H_L = X_L, \quad H_L X_L H_L = Z_L.$$

12.05 · Joint parity and entanglement

EXPLANATION

A joint parity measurement reveals whether two logical values agree. It preserves coherence between alternatives with the same parity, rather than resolving both individual values.

Starting with two logical plus states, either parity outcome produces an entangled Bell state after normalization. The outcome identifies which Bell state was obtained, so it must be retained for later interpretation.

Lattice surgery implements such logical parity measurements through temporary changes of local boundary checks. Repeated rounds protect the parity inference. The ideal projector states the target measurement; a concrete surgery needs its own merge, split, measurement, and decoding schedule.

$$M_{ZZ} = \bar{Z}_A \bar{Z}_B, \quad \Pi_m^{ZZ} = \frac{I + (-1)^m M_{ZZ}}{2}, \quad m \in \{0, 1\}.$$

$$p_m = \text{Tr}(\Pi_m^{ZZ} \rho), \quad \rho_m = \Pi_m^{ZZ} \rho \Pi_m^{ZZ} / p_m.$$

For $|+_L\rangle_A |+_L\rangle_B$:

$$\Pi_0^{ZZ} |+_L, +_L\rangle = (|0_L 0_L\rangle + |1_L 1_L\rangle)/2,$$

$$\Pi_1^{ZZ} |+_L, +_L\rangle = (|0_L 1_L\rangle + |1_L 0_L\rangle)/2, \quad p_0 = p_1 = 1/2.$$

$$\Pi_m^{XX} = \frac{I + (-1)^m \bar{X}_A \bar{X}_B}{2}.$$

12.06 · Entangling gates and Clifford structure

EXPLANATION

A logical CNOT has the same basis action and Pauli propagation rules as an ordinary CNOT, but acts on encoded qubits. Surgery protocols can realize it using joint parity measurements, an ancilla patch, and outcome-dependent corrections.

The conjugation rules provide a useful algebraic test of the intended Clifford action. They do not replace verification of the fault-tolerant measurement schedule.

Clifford operations preserve the Pauli group under conjugation, which makes frame tracking efficient. The T gate takes a Pauli outside that group, so a universal gate set requires additional resources beyond these Clifford operations.

MATHS

Logical CNOT:

$$\text{CNOT}_L |a_L, b_L\rangle = |a_L, (a \oplus b)_L\rangle.$$

$$\bar{X}_A \mapsto \bar{X}_A \bar{X}_B, \quad \bar{Z}_A \mapsto \bar{Z}_A, \quad \bar{X}_B \mapsto \bar{X}_B, \quad \bar{Z}_B \mapsto \bar{Z}_A \bar{Z}_B.$$

$$\text{Cliff}_k = \{U : U \mathcal{P}_k U^\dagger = \mathcal{P}_k\}, \quad H, S, \text{CNOT} \text{ generate Clifford operations (up to phase).}$$

$$S = \text{diag}(1, i), \quad T = \text{diag}(1, e^{i\pi/4}), \quad TXT^\dagger = (X + Y)/\sqrt{2} \notin \mathcal{P}_1.$$

12.07 · Ideal magic-state injection

EXPLANATION

An ideal magic state can be consumed to implement T using a CNOT, a measurement, and a conditional S correction. In the convention shown, the data qubit controls the CNOT and the magic-state ancilla is measured in Z.

One measurement branch applies T directly. The other applies its inverse up to phase, and the S correction converts it into T. This derivation states the circuit orientation so the feed-forward rule is unambiguous.

For protected surface-code computation, preparing the required encoded resource with sufficiently low error is a separate task. Injection and typically distillation address that task. The ideal circuit is not a proof that a noisy injected state produces a fault-tolerant T gate, and the conditional correction is Clifford rather than merely Pauli.

MATHS

All qubits and gates below may be interpreted at the ideal logical level.

$$|A\rangle = T|+\rangle = (|0\rangle + e^{i\pi/4}|1\rangle)/\sqrt{2}.$$

Input $|\psi\rangle_d |A\rangle_a$; apply $\text{CNOT}_{d \rightarrow a}$; measure ancilla Z with bit m :

$$K_m = {}_a\langle m | \text{CNOT}_{d \rightarrow a} | A \rangle_a.$$

$$K_0 = \frac{T}{\sqrt{2}}, \quad K_1 = \frac{e^{i\pi/4}T^\dagger}{\sqrt{2}}, \quad SK_1 = \frac{e^{i\pi/4}T}{\sqrt{2}}.$$

$$F_0 = I, \quad F_1 = S, \quad F_m K_m |\psi\rangle \sim \frac{T|\psi\rangle}{\sqrt{2}}.$$

12.08 · Exercise and result

EXPLANATION

Explain the three requirements separately. Code preservation says valid inputs stay within a valid code space. Logical correctness says the transformation inside that space is the intended one. Fault tolerance says the implementation remains reliable under the declared physical faults.

A logical Z preserves the code space perfectly but would be wrong if the requested operation were identity. Conversely, an ideal circuit can implement the right gate while allowing one ancilla fault to spread into an uncorrectable error. These examples show why each requirement needs its own check.

MATHS

Code preservation: $U_{\text{phys}} P U_{\text{phys}}^\dagger = P$.

Specified logical action: $U_{\text{phys}} V = V U_L$.

Fault tolerance: specified allowed faults remain correctable through the gadget.

$$\bar{Z} P \bar{Z}^\dagger = P, \quad \bar{Z} V = V Z \text{ / } V I_L =$$